PROCESS CELL: PC_CHI_BAKE — CHI Baking Oven Process Cell

## EQUIPMENT LIST — PC_CHI_BAKE

| EXT ID | NAME | TYPE | MANUFACTURER | PARENT EM |
|---|---|---|---|---|

## BAKING ZONE — OVEN 1

**EQ_AUX_003**
Temp Sensor 3 [RTD]

**EQ_AUX_017**
Temp Sensor 17 [RTD]

TT 011 — TEMP_DEGC
TDT 012 — ZONE_DELTA_DEGC
FT 013 — AIRFLOW_M3H
HT 014 — HUMIDITY_PCT
TT 010 — EXHAUST_TEMP_DEGC

FUEL GAS
EXHAUST
COMB. AIR

### EQ_CHI_014
**CHI Oven 1**
*(Tunnel Oven)*

OS 030 — OPERATING_STATE
AS 031 — ALARM_STATE

Mfr: Rockwell Automation

PT 015 — PRESSURE_BAR
BT 016 — BURNER_LOAD_PCT
ST 017 — SPEED_MPM
IT 018 — MOTOR_CURRENT_A
FWT 019 — THROUGHPUT_KG_H

*PLC CABINET — UNIT_CHI_BAKE / EM_CHI_BAKE*

- **CM_COOKIE_001** — Oven Zone — Control 1 — *[PID]* — PLC-01/DB1200
- **CM_COOKIE_002** — Oven Burner — Mgmt 1 — *[SEQ]* — PLC-02/DB1201
- **CM_COOKIE_009** — Oven Zone — Control 2 — *[PID]* — PLC-09/DB1208
- **CM_COOKIE_010** — Oven Burner — Mgmt 2 — *[SEQ]* — PLC-10/DB1209

**FROM PC_CHI_FORM**
*(FORMED DOUGH)*

DOUGH TRANSFER BELT

## BAKING ZONE — OVEN 2

**EQ_AUX_024**
Temp Sensor 24 [RTD]

TT 021 — TEMP_DEGC
TDT 022 — ZONE_DELTA_DEGC
FT 023 — AIRFLOW_M3H
HT 024 — HUMIDITY_PCT
TT 020 — EXHAUST_TEMP_DEGC

FUEL GAS
EXHAUST
COMB. AIR

### EQ_CHI_015
**CHI Oven 2**
*(Tunnel Oven)*

OS 030 — OPERATING_STATE
AS 031 — ALARM_STATE

Mfr: GEA

PT 025 — PRESSURE_BAR
BT 026 — BURNER_LOAD_PCT
ST 027 — SPEED_MPM
IT 028 — MOTOR_CURRENT_A
FWT 029 — THROUGHPUT_KG_H

*PLC CABINET — UNIT_CHI_BAKE / EM_CHI_BAKE*

- **CM_COOKIE_017** — Oven Zone — Control 3 — *[PID]* — PLC-01/DB1216
- **CM_COOKIE_018** — Oven Burner — Mgmt 3 — *[BATCH]* — PLC-02/DB1217
- **CM_COOKIE_025** — Oven Zone — Control 4 — *[PID]* — PLC-09/DB1224
- **CM_COOKIE_026** — Oven Burner — Mgmt 4 — *[SEQ]* — PLC-10/DB1225

**TO PC_CHI_COOL**
*(BAKED PRODUCT)*

### LEGEND

- TT — Temperature Transmitter
- TDT — Zone ΔTemp Transmitter
- FT — Airflow Transmitter
- PT — Pressure Transmitter
- HT — Humidity Transmitter
- BT — Burner Load Transmitter
- ST — Speed / Belt Transmitter
- IT — Motor Current Indicator
- FWT — Throughput Transmitter
- OS — Operating State
- AS — Alarm State

**NOTES:**

1. All instrument tags reference ISATimeSeries external IDs (isString = false / numeric).
2. CMs CM_COOKIE_001/002/009/010/017/018/025/026 assigned to EM_CHI_BAKE.
3. EQ_CHI_014 (Oven 1, Rockwell) and EQ_CHI_015 (Oven 2, GEA) run in series.
4. State encoding: 0=STOPPED 1=RUNNING 2=IDLE 3=STARTUP 4=ALARM 5=MAINTENANCE.
5. Hierarchy: PC_CHI_BAKE → UNIT_CHI_BAKE → EM_CHI_BAKE.
6. Fuel gas / combustion air / exhaust utility connections not fully instrumented here.